Ultrasound Lets Scientists Check 'Pregnant' Coral Without Harming It

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Coral can now get a checkup without being cut open. Scientists have used an **ultrasound** machine to look inside living coral underwater and spot eggs developing within it. It is the first time researchers can assess a coral's reproductive health without harming or **sacrificing** the animal.

Until now, checking whether coral was carrying eggs usually meant cutting it open. The new approach uses sound waves instead. Canon, the camera maker, announced this week that its advanced medical **ultrasound** platform, the Aplio i700, can scan coral submerged underwater. With the United Kingdom conservation group Coral **Spawning** International, the team scanned six species of large **polyp** stony coral. Its high and ultra-high frequency probes let researchers see through the corals' skeletons.

To confirm the scans, the team cut open one coral species, Trachyphyllia geoffroyi, and took high-powered images of its interior. The images matched the **ultrasound** results, confirming that the structures seen in the scans were indeed eggs.

Knowing when corals will release their eggs matters. It can help **conservationists** predict **spawning** and time their restoration work, giving breeding programs a head start. **Ultrasound** scans could make that work more precise while sparing corals from being cut apart.

Coral needs all the help it can get. Late last month, the International Union for Conservation of Nature warned that nearly all of Earth's coral reef ecosystems, 96 percent, face an extremely high risk of collapse within the next 50 years, driven by warming oceans and heat waves. Reefs, sometimes called the rainforests of the sea, support an estimated quarter of all marine species. When reefs collapse, coral dies and is replaced by algae and sponges that support far less marine life.

Word Treasure

ultrasound -- A scan that uses sound waves to make pictures of the inside of something.

polyp -- A tiny soft animal that builds a coral's hard skeleton.

spawning -- When corals release their eggs into the water.

conservationists -- People who work to protect nature and wildlife.

sacrificing -- Giving something up or destroying it for a purpose.

Background you need

Coral is a living animal that builds a hard skeleton, and many corals release eggs into the water so new corals can grow.

An ultrasound machine sends sound waves into the body and builds a picture from the echoes, which lets doctors look inside a person without surgery.

Scientists are now using that same idea underwater to see eggs growing inside living coral, so they can tell when the coral is ready to spawn.

Knowing when spawning will happen helps conservationists plan their work to protect and restore reefs, which support many kinds of ocean life.

Break it down

LEAD: Scientists used an ultrasound machine to spot eggs inside living coral without harming it.
KEY EVENT: Canon and Coral Spawning International scanned six species of large polyp stony coral with the Aplio i700.
+- WHO: Canon, Coral Spawning International, and researchers
+- HOW: High and ultra-high frequency probes sent sound waves through the corals' skeletons
+- CHECK: The team cut open Trachyphyllia geoffroyi and high-powered images matched the ultrasound scans
EVIDENCE: The images confirmed the structures seen in the scans were indeed eggs.
WHY IT MATTERS: Predicting spawning helps time restoration work, and reefs support about a quarter of all marine species.

Why it matters

Reefs support about a quarter of all marine species, and nearly all reef ecosystems face an extremely high risk of collapse, so a gentle way to track coral reproduction can help conservationists protect and rebuild them.

Test what you remember

1. **What did scientists use to look inside living coral without harming it?**
   - (A) An X-ray machine
   - (B) A microscope
   - (C) An ultrasound machine
   - (D) A camera flash
2. **How many species of large polyp stony coral did the team scan?**
   - (A) Ten
   - (B) Six
   - (C) Two
   - (D) Three
3. **What did the team do to confirm that the ultrasound scans were correct?**
   - (A) They cut open one coral species and took high-powered images of its interior
   - (B) They waited for the coral to spawn
   - (C) They asked other scientists to guess what the scans showed
   - (D) They scanned the coral a second time
4. **Which coral species did the team cut open to check the scans?**
   - (A) International Union for Conservation of Nature
   - (B) Trachyphyllia geoffroyi
   - (C) Aplio i700
   - (D) Coral Spawning International
5. **According to the warning from the International Union for Conservation of Nature, what share of Earth's coral reef ecosystems face an extremely high risk of collapse within the next 50 years?**
   - (A) 50 percent
   - (B) 25 percent
   - (C) 10 percent
   - (D) 96 percent
6. **Why does knowing when corals will release their eggs matter?**
   - (A) It lets scientists count every fish on a reef
   - (B) It helps conservationists predict spawning and time their restoration work
   - (C) It makes coral grow faster
   - (D) It stops the oceans from warming

Answer key (try the quiz first!): 1: C 2: B 3: A 4: B 5: D 6: B

Questions to think about

MY THOUGHTS (at least 20 words):